EMERGENCY EVACUATION FLOW OPTIMIZER USING THE EDMONDS–KARP ALGORITHM

BFS-Based Maximum Flow Algorithm Visualization

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ABSTRACT

- Develops a web-based application to analyze evacuation flow during emergencies such as fires and floods.
- Models evacuation networks using nodes for locations and directed edges for routes with limited capacities.
- Uses the Edmonds–Karp algorithm with BFS to identify augmenting paths.
- Visualizes flow updates, bottleneck capacities, and residual graph changes at each iteration.
- Calculates the maximum possible evacuation flow from a danger zone to a safe zone.
- Demonstrates the practical application of graph algorithms in emergency management.

PROBLEM STATEMENT

Urban road networks have limited vehicle-handling capacities, and traffic flow between junctions can be restricted by bottlenecks and limited alternative routes. Identifying the maximum possible flow through a network while respecting individual road capacities is a challenging network optimization problem. Therefore, an effective approach is needed to model road networks and analyse their flow capacities to support better traffic planning and resource utilization.

PROJECT OBJECTIVE

- To model an urban road network using intersections as nodes and roads as directed edges with defined capacities.
- To determine the maximum possible traffic flow between a source junction and a destination junction using the Edmonds–Karp algorithm.
- To identify bottlenecks and visualize BFS-based augmenting paths and residual capacity updates at each iteration.
- To provide an interactive visualization that improves understanding of maximum-flow analysis and supports traffic network planning.

DSA CONCEPTS USED

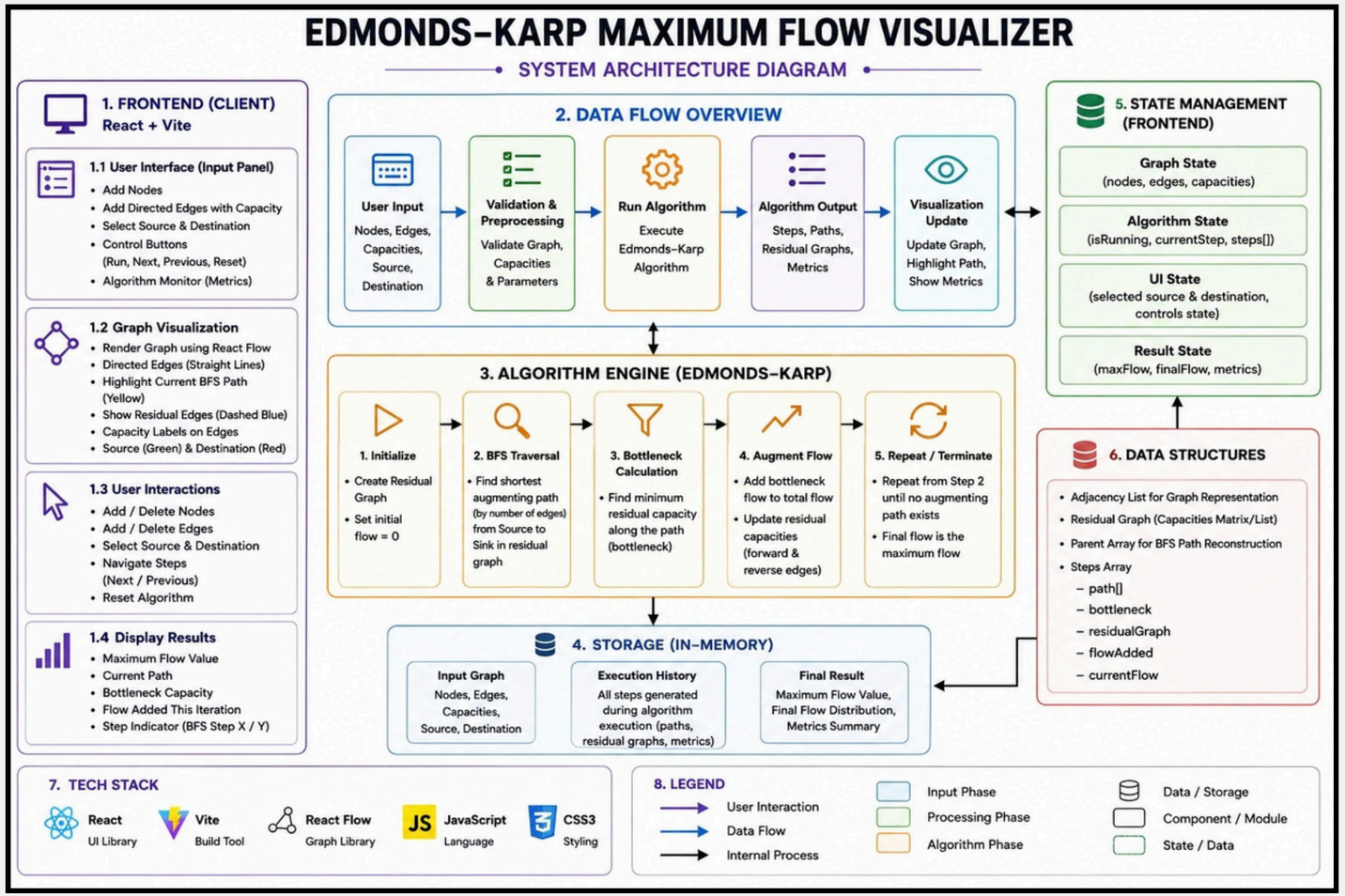
- Graph
  --Directed Graph
  --Adjacency representation
- Queue
  --Used in BFS traversal
- BFS
  --Finds the shortest augmenting path
- Residual Graph
  --Stores remaining capacities after flow updates
- Edmonds–Karp
  --Repeatedly applies BFS to find augmenting paths
  --Updates residual capacities
  --Calculates maximum flow

TOOLS/TECHNOLOGIES USED

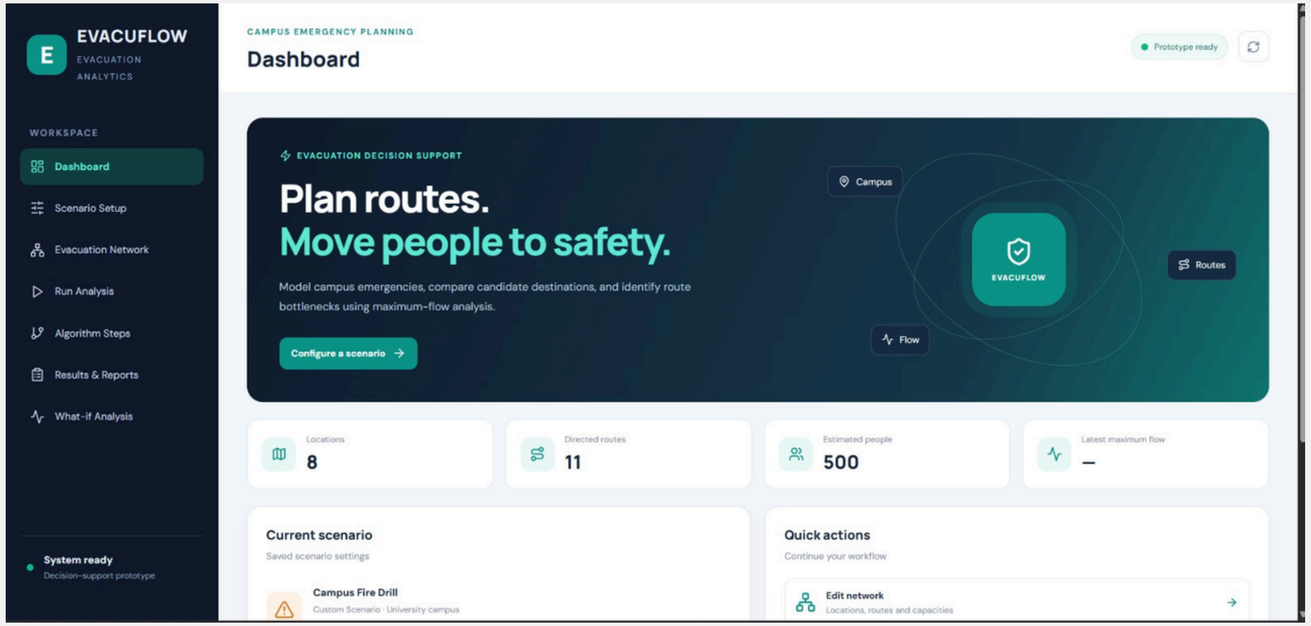
- Frontend
  --React
  --JavaScript
  --HTML
  --CSS
- Development
  --Vite
- Graph Visualization
  --React Flow
- Algorithm

Edmonds–Karp
  --Breadth-First Search
  --Residual Graph

SYSTEM ARCHITECTURE



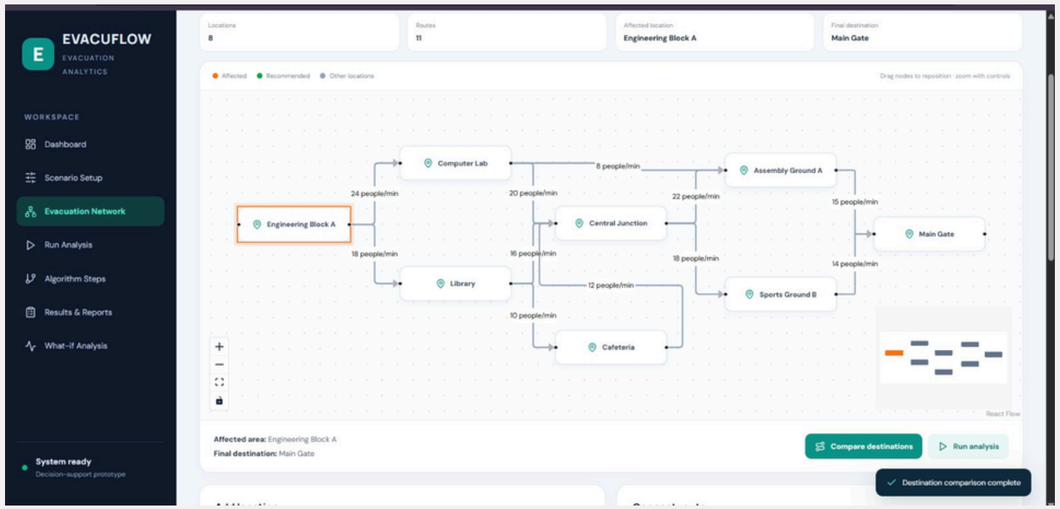
FRONTEND IMPLEMENTATION



Implemented Features

- Displays locations, routes, and estimated people.
- Provides quick access to all modules.
- Shows current evacuation scenario details.

ALGORITHM VISUALIZATION

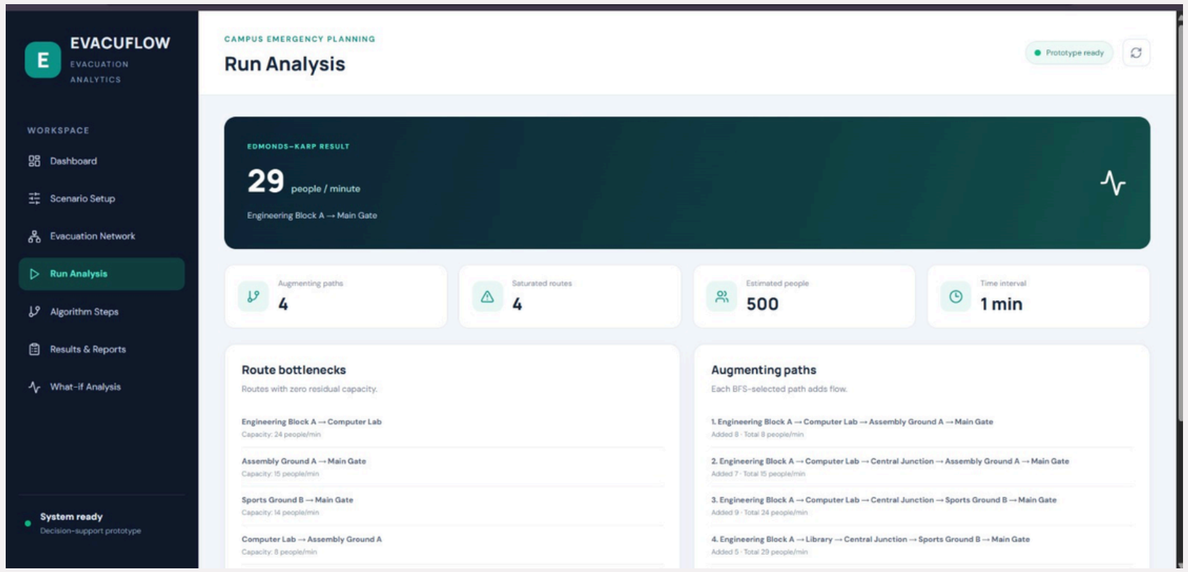


Implemented Features

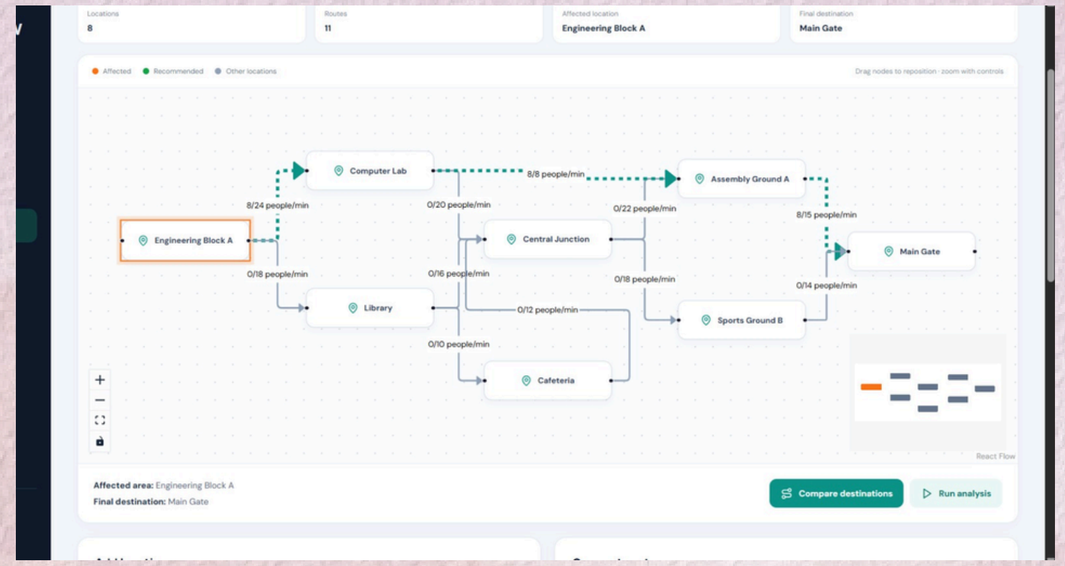
- Visualizes locations and directed routes.
- Displays route capacities and evacuation paths.
- Highlights affected locations and destinations.

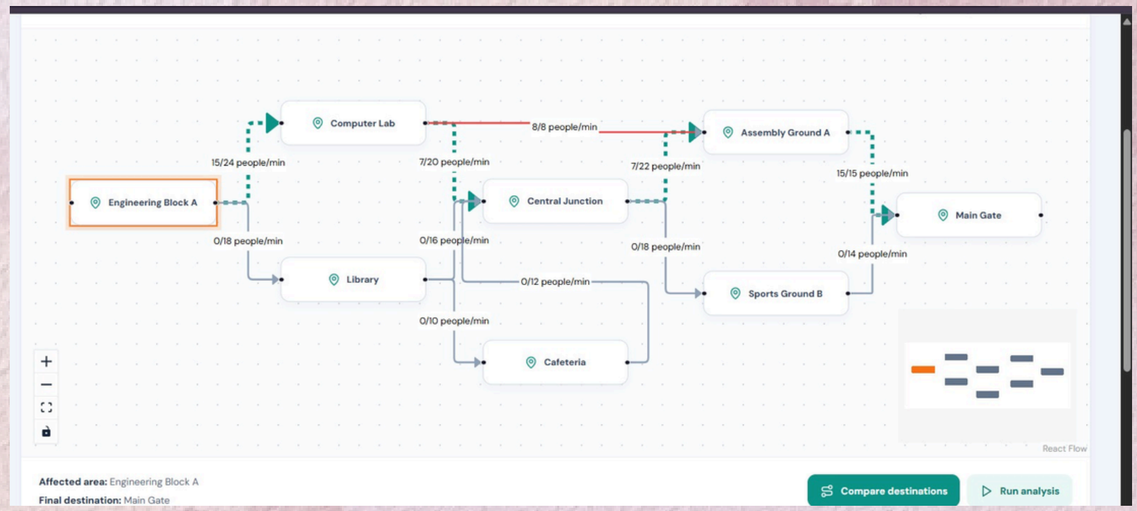
Implemented Features

- Displays the calculated maximum flow.
- Summarizes augmenting paths and saturated routes.
- Supports printing and saving reports as PDF.



ALGORITHM VISUALIZATION





FUTURE SCOPE

- Integrate real-time traffic data to analyse changing road conditions and vehicle flow.
- Extend the system to identify traffic bottlenecks and evaluate alternative routes across larger road networks.
- Incorporate dynamic road capacities to represent changing traffic conditions during peak and off-peak hours.
- Develop an interactive map-based interface to model real-world road networks and support traffic planning decisions.

CONCLUSION

The project successfully demonstrates how graph algorithms can be applied to real-world traffic network problems. By implementing the Edmonds–Karp algorithm, our website provides a visual representation of how road capacities influence traffic flow and how bottlenecks affect the movement of vehicles through a network. The system combines algorithmic problem-solving with interactive visualization, making maximum-flow analysis easier to understand and providing a basis for future improvements in traffic network modelling.

THANK YOU